Predicting Heart Disease from Clinical Indicators

A reproducible logistic regression prototype

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The Problem

Binary classification: does this patient have heart disease?

Context

Patients present with chest pain. A predictive model helps triage those likely to have coronary heart disease.

Data: UCI multi-study dataset (920 patients, 4 source studies, 14 clinical features).

Target: binary indicator derived from num > 0.

Approach

1. Explore data, flag quality issues
2. Select features, handle missingness
3. Build reproducible sklearn pipeline
4. Evaluate on CV + hold-out test
5. Interpret + recommend next steps

920

patients

across 4 studies

55%

disease

overall prevalence

14

features

clinical + demographic

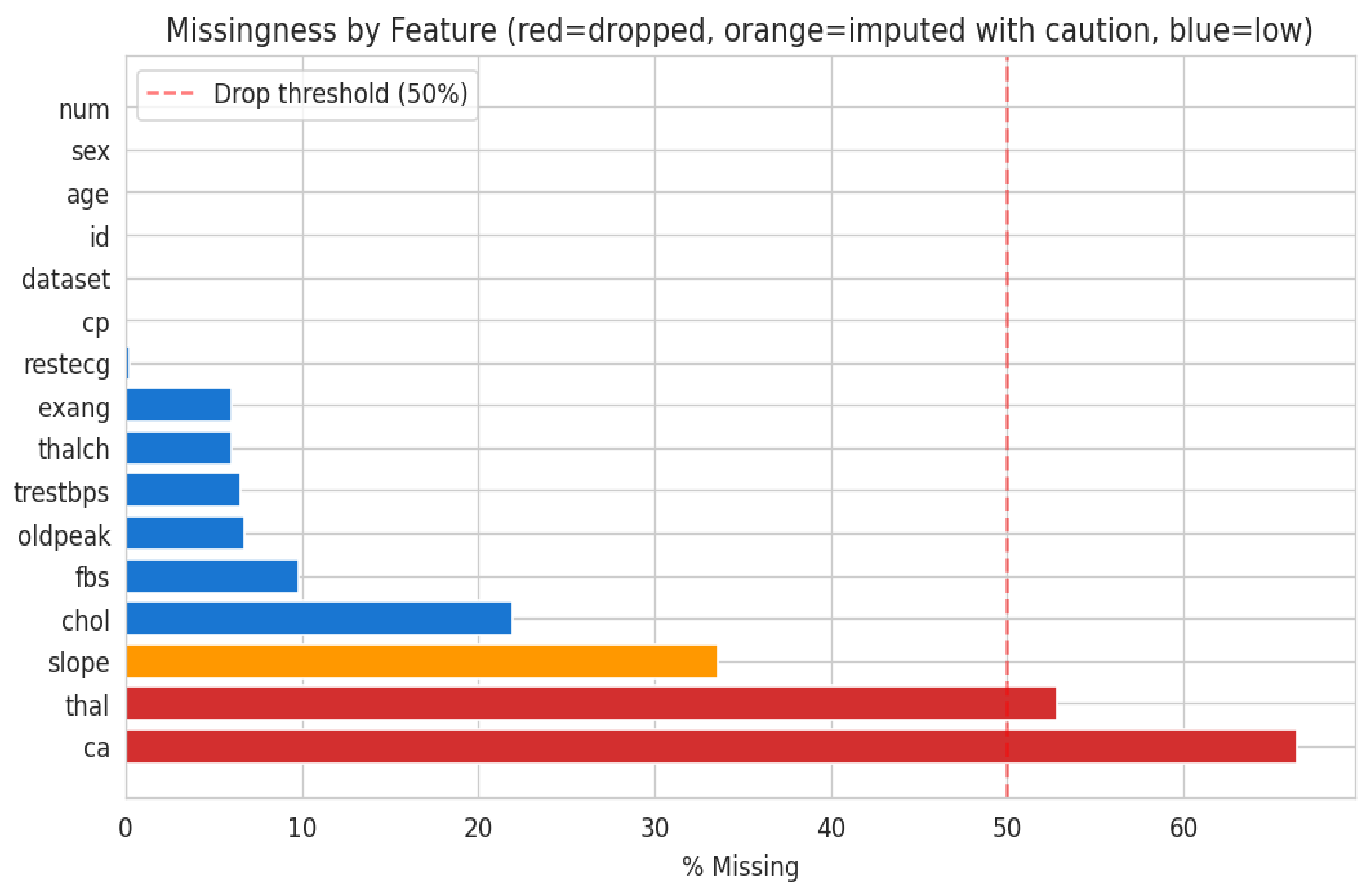
Guiding Principle

"Prototype over precision":

Working end-to-end pipeline with defensible choices, not leaderboard accuracy. Modular RAP-compliant project: every decision traceable, every result reproducible (seed = 42).

Data Exploration (1 of 2)

Missingness and data quality drive early design choices



Key findings

ca: 66% missing
number of major vessels by fluoroscopy

thal: 53% missing
thalassemia result

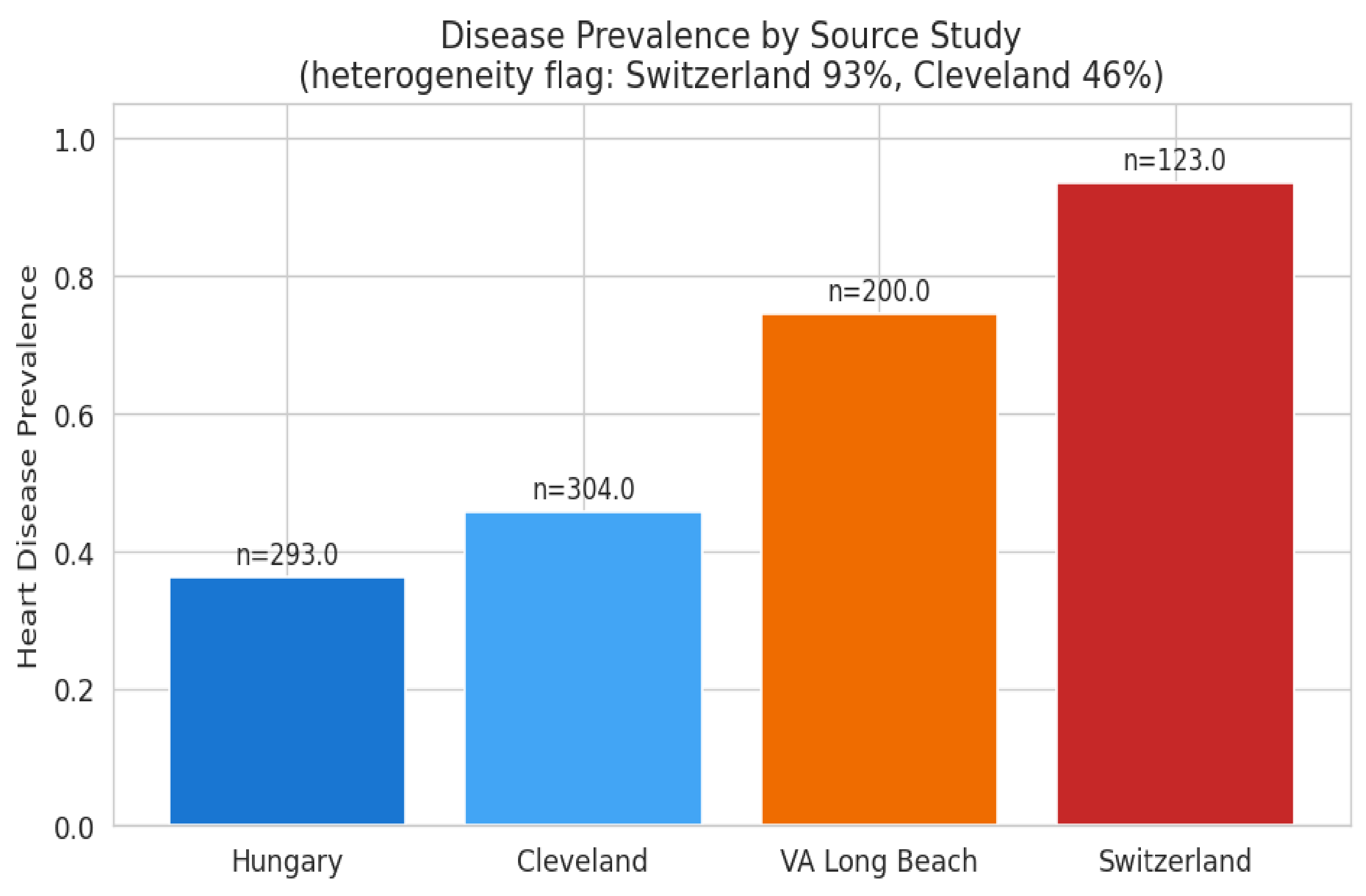
slope: 34% missing
ST segment slope at peak exercise

Other features <10% missing

Data quality issue:
trestbps = 0 and chol = 0 in 172 records — physiologically impossible, recoded as missing.

Data Exploration (2 of 2)

Source-study heterogeneity is the biggest analytical risk



Why this matters

Prevalence varies 36% to 94% across studies.

Likely reasons:

- selection effects (referral pathway)
- diagnostic protocol differences
- age/sex case-mix variation

Risk: confounding by study site.

If ignored, the model may learn 'Switzerland' as a disease proxy rather than true clinical signal.

My mitigation: include dataset as a feature so the model conditions on study site explicitly.

Design Decisions

Four choices that define the model

Feature set

Drop ca and thal (>50% missing). Keep slope with imputation. Include dataset as feature.

Why: Imputation above ~50% is unreliable. Dropping 2 features costs less than introducing noise.

Missing data

Median imputation for numeric, mode for categorical, inside a sklearn Pipeline.

Why: Embedded imputation prevents train/test leakage. Median is robust to outliers.

Model choice

Logistic regression with L2 regularisation.

Why: Interpretable coefficients, clinical standard, suits 736 training rows, strong baseline before tree methods.

Evaluation

80/20 stratified split + 5-fold CV on train. Metrics: ROC-AUC, recall, precision.

Why: Recall prioritised — false negatives (missed diagnoses) carry more clinical risk than false alarms.

Pipeline Architecture

Modular RAP-compliant project, single-command reproducible

Project structure

```
heart_disease_rap/
  config.yaml            # all params
  main.py                # orchestrator
  requirements.txt
  README.md   RAP.md
  src/
    config.py
    data_loader.py       # stage 1
    eda.py               # stage 2
    preprocess.py        # stage 3
    train.py             # stages 4-6
    evaluate.py          # stage 7
  tests/
    test_pipeline.py     # 4 tests
  data/raw/
  outputs/               # generated
```

RAP principles satisfied

Version controlled
Git + config.yaml; every run logged with seed and metrics in run_summary.json.

Modular and reusable
Seven single-purpose modules. Preprocessor, data loader, evaluator are independently reusable.

Automated and orchestrated
Single command: python main.py. No notebook cells, no manual steps.

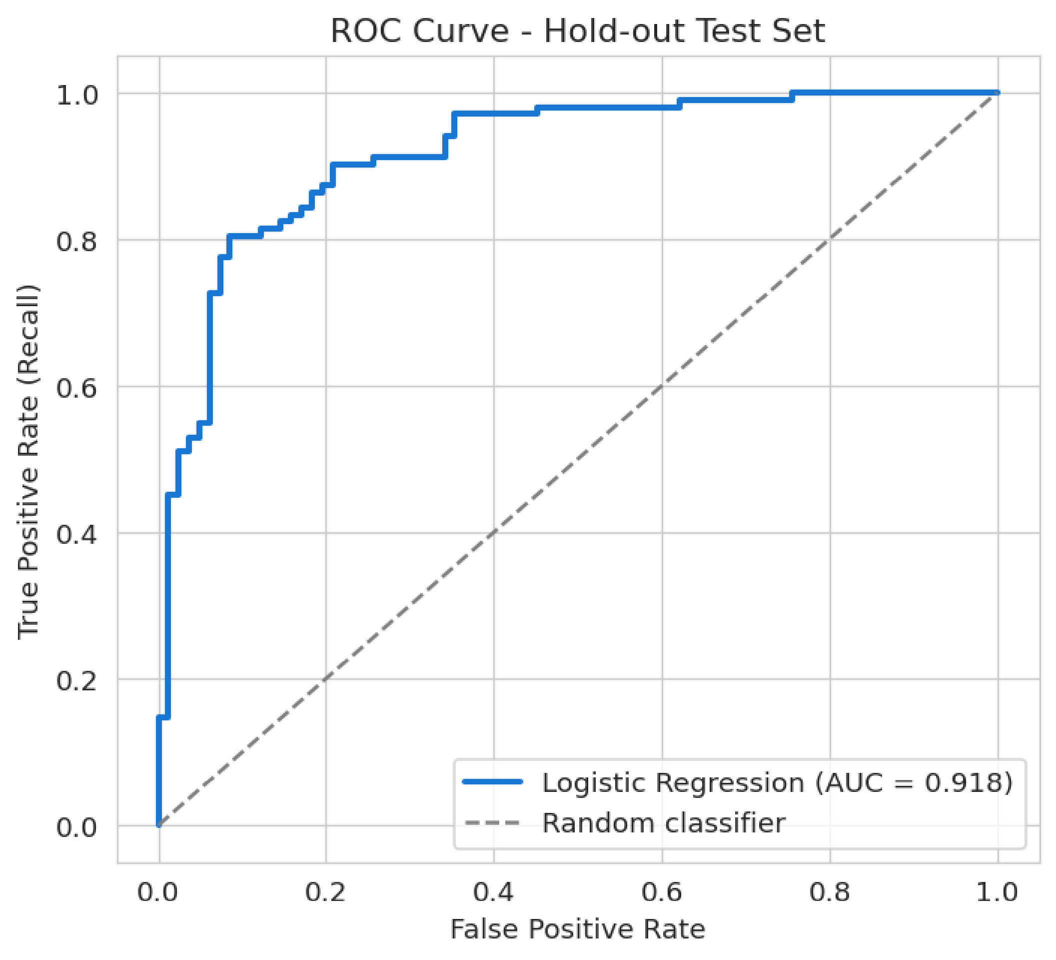
Documented
README, RAP.md, module docstrings, function type hints, structured logging.

Tested
Four pytest cases: DQ, target derivation, preprocessor, end-to-end smoke test.

Open source and reproducible
No proprietary tools. Fixed seed. Identical metrics on every run.

Model Performance

Strong discrimination, recall prioritised



CV ROC-AUC

0.890

+/- 0.019 over 5 folds

Test ROC-AUC

0.918

held-out 184 patients

Recall

0.902

92/102 disease cases caught

Precision

0.829

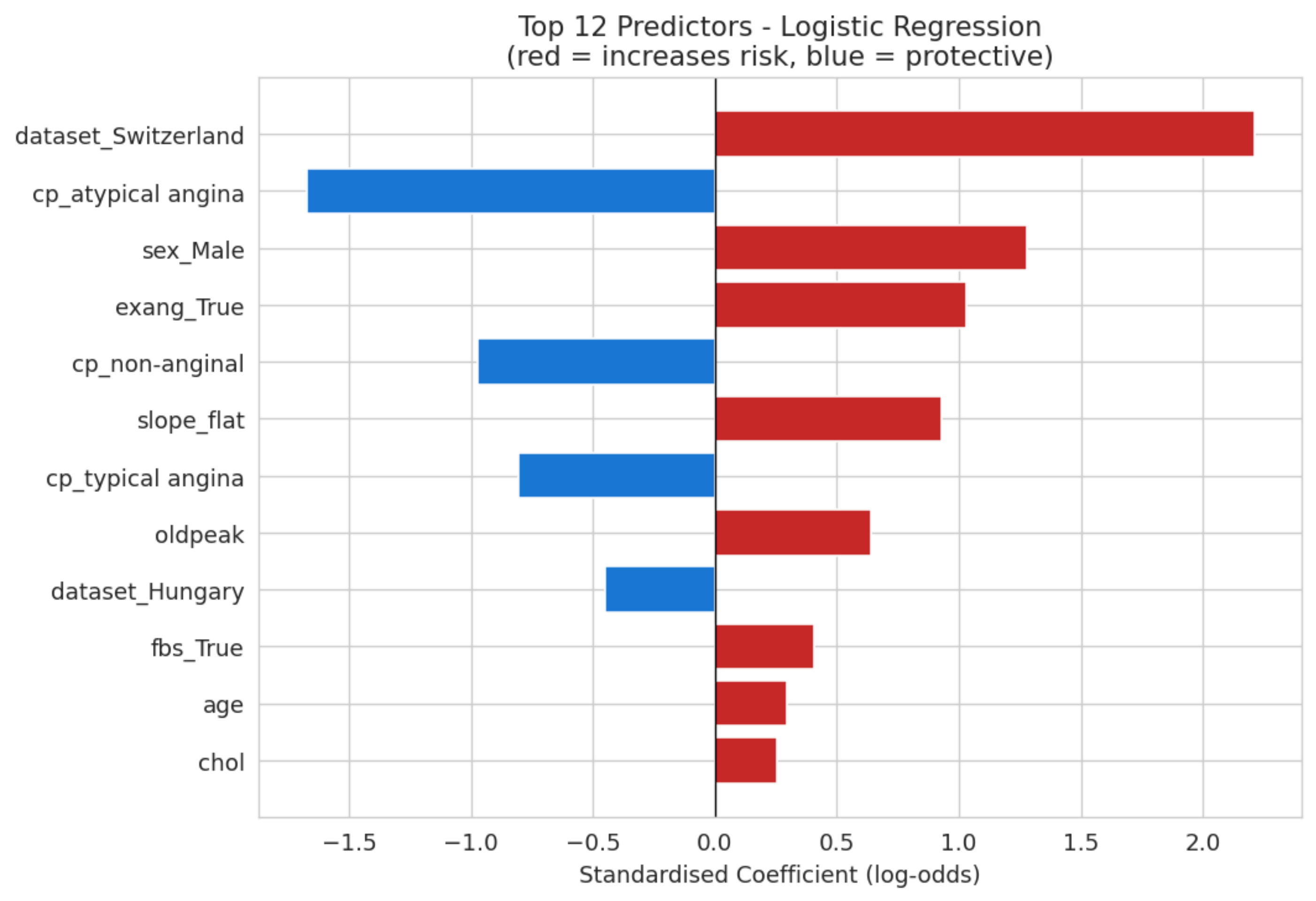
92/111 flagged truly diseased

Confusion matrix (test set, n=184)

	Predicted: no disease	Predicted: disease
Actual: no disease	63 (TN)	19 (FP)
Actual: disease	10 (FN)	92 (TP)

Interpretation

The model's top predictors are clinically plausible



Does this make clinical sense?

Clinically consistent:

Male sex: higher risk
Exercise-induced angina: strong positive
Atypical / non-anginal chest pain: protective
ST depression (oldpeak), age: positive

Flag: dataset_Switzerland dominates

Strongest coefficient (+2.2). Driven by 94% disease prevalence in Swiss subset, not clinical signal.

Interpretation rule:

Trust the clinical-feature coefficients. Treat the dataset coefficients as case-mix adjustment, not as causal drivers.

Limitations and Next Steps

How to take this prototype forward

Methodological

- Test tree-based models (RF, gradient boosting) for non-linear signal.
- Formal study-stratified analysis to separate clinical from site effect.
- Calibration analysis + Brier score, not just discrimination.
- Fairness audit by sex and age group.

Data

- Recover ca and thal where clinically feasible.
- External validation on UK NHS patient data.
- Link with outcome data (MI, mortality).
- Expand features: medication, family history, comorbidity.

Operationalisation

- Package pipeline behind an API for clinical tool integration.
- MLOps: monitoring, drift detection, retraining schedule.
- AQA / model-governance sign-off before any real-world use.
- Clinician-facing UI with explainability (SHAP values).

Summary

A reproducible logistic regression prototype delivering 0.92 ROC-AUC on held-out test data.

Designed to be defended, not just deployed.

- End-to-end pipeline, seed-controlled, runs in one command.
- Explicit handling of missingness, data errors, and study heterogeneity.
- Recall 0.90 prioritised — clinically appropriate for screening.
- Clear roadmap for methodological, data, and operational next steps.

Questions welcome.